SRCC GBO 2013 Question Paper

TAKE THIS MOCK IN EXAM FORMAT

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Data Interpretation

Instructions [1 - 5]

The table given below indicates the number of consumers who preferred one or both drinks. Read the table and answer questions.

Age of Consumer (in year)	Drink preferred						Total number of Consumers surveyed including those who did not prefer any drink	
	A		B		Both			
	Male	Female	Male	Female	Male	Female	Male	Female
18-25	240	180	180	120	90	60	600	720
26-40	360	120	480	360	90	180	840	540
41-60	300	240	240	300	120	180	1200	960

1. Number of consumers in the age group 26-40 who did not prefer drink B is

A 540

B 480

C 360

D 180

2. Number of consumers over 40 years who preferred at least one drink is

A 450

B 780

C 870

D 1320

3. The number of consumers in the age group 18-25 who prefer only one drink is

A 1020

B 720

C 570

D 420

4. The percent of consumers more than 40 years of age who did not prefer any one of the drink is about

A 43.2

B 54.7

C 63.9

D 76.1

5. In all, how many consumers did not prefer any drink?

A 1080

B 1380

C 2160

D 2460

Instructions [6 - 10]

The following table classifies the tea and coffee habits of all employees of a company. Complete the table and answer the questions.

Tea			
	Less than 2 cups a day	2 to 3 cups a day	More than 3 cups a day
Males	95	?	?
Females	88	45	67
Total	?	106	?

Cofee			
	Less than 2 cups a day	2 to 3 cups a day	More than 3 cups a day
Males	85	?	49
Females	40	108	?
Total	?	144	101

6. How many Employees drink more than 3 three cups of tea a day?

A 183

B 106

C 81

D 67

7. Number of females who drink two or more cups of a tea per day is

A 45

B 108

C 112

D 160

8. The percentage of male employees in the company is close to

A 43.5

B 45.9

C 47.8

D 54.1

9. The number of employees who drink upto three cups of coffee per day is

A 269

B 264

C 221

D 156

10. The ratio of the number of males who drink two or more cups of tea per day to the number of employees who drink less than 2 cups of coffee per day is

A 15:17

B 19:25

C 5:3

D 3:5

Instructions [11 - 15]

The following table gives demand and supply of cement, in million tons, for the period 2005 to 2010. Surplus is defined as excess of supply over demand. Read the table and answer the Questions.

Year	Demand	Supply
2005	45.1	48.2
2006	47.7	48.3
2007	50.0	51.2
2008	53.4	54.2
2009	57.5	60.0
2010	61.7	62.4

11. The surplus of cement was lowest for the year

A 2006

B 2007

C 2008

D 2010

12. The percentage increase in demand of cement was the highest as compared to its previous year in

A 2006

B 2007

C 2009

D 2010

13. What was the approximate average surplus (in million tons) of cement for the period 2005 to 2010 ?

A 2001

B 2002

C 2003

D 2005

14. In how many years the supply (in million tons) of cement was more than the average supply for 2005 to 2010 ?

A 4

B 3

C 2

D 1

15. For the years 2005 to 2010, total supply of cement was x % more thanthe total demand. The value of x is closest to

A 1.8

B 2.1

C 2.3

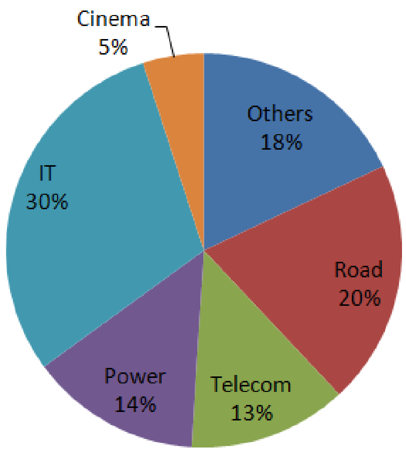
D 2.8

Instructions [16 - 20]

The following table shows the FDI in different states of a country in the year 2009-10.

State	A	B	C	D	E	F	G
FDI (in ₹ crores	550	580	660	560	480	510	600

The following graph shows the investment in different sectors.



Read the table and graph and answer the questions.

16. In which of the following pairs of states the ratio of investment (FDI) in power sector is 11 : 8 ?

A A and B

B B and C

C C and E

D A, D

17. The ratio of investment in IT in state G to the investment in 'others' in state A is

A 11:20

B 20: 11

C 5:3

D 3:5

18. The total investment (in ₹ crore) in 'Road Sector' by states B, C and D is

A 360

B 340

C 320

D 398

19. The FDI in cinema sector in state E is about what percent less than that in state B in telecom sector ?

A 47

B 58

C 63

D 68

20. FDI in IT sector in State A is approximately what percent of that in telecom sector in state G and F combined ?

A 114.3

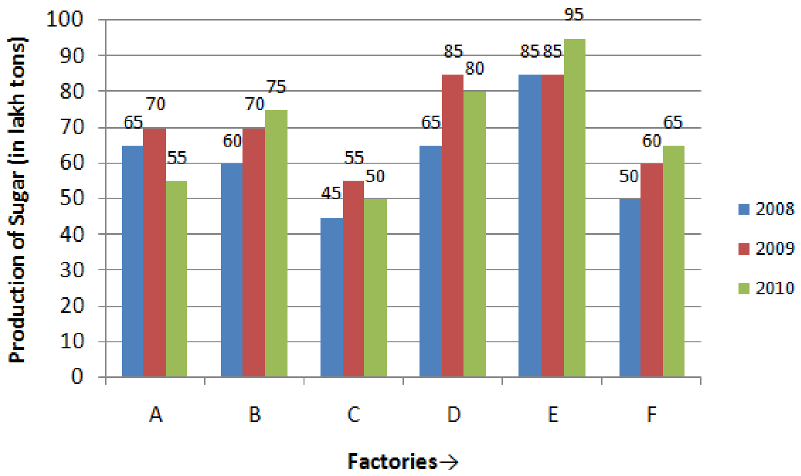
B 118.3

C 121.7

D 138.5

Instructions [21 - 25]

Read the following graph and answer the questions:



21. The difference (in lakh tons) between the average production of sugar by six factories in 2009 and average production by the same factories in 2008 is about

A 6.6

B 9.2

C 9.4

D 9.8

22. The percentage increase in production of sugar by factory B from 2009 to 2010 is approximately

A 11.7

B 8.3

C 7.1

D 1.6

23. Which of the six factories has recorded the maximum percentage growth in production from 2008 to 2009 ?

A B

B C

C D

D E

24. Production of sugar by factory C in 2009 and production of sugar by factory F in 2010 together is what percent of production by B in 2010 ?

A 160

B 150

C 133.3

D 08.4

25. In which of the following pairs of factories, the difference between average production for 2009 and 2010 is minimum?

A A and B

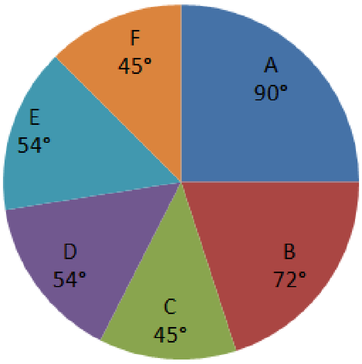
B B and c

C B and D

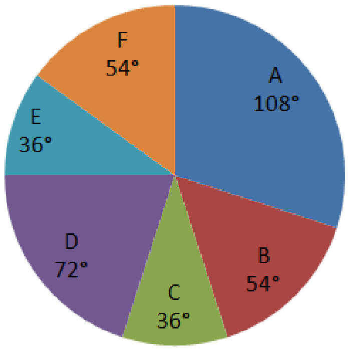
D D and E

Instructions [26 - 30]

The following pie-charts show the distribution of expenses over six different items A, B, C, D, E and F of a person for July and August 2012. She spent ₹ 27,000 in July and ₹ 30,000 in August, 2012



August, 2012



July, 2012

Read the charts and answer the questions

26. The ratio of money spent one items C and F together in July to that spent on B and E together in August is

A 9:10

B 10:9

C 3:4

D 4:5

27. The difference (in ₹) between the amounts spent on C in August and D in July is

A 1,250

B 1,125

C 1,050

D 900

28. The sum of the difference between the expenditurers in july and August on items C and that of F as a percentage of the change in total expenditure between the two months is

A 25

B 37.5

C 47.5

D 50

29. Number of items of July on which money spent is less than the average money spent on items in the month is

A 5

B 4

C 3

D 2

30. If the amount spent on item B in August were the same as that in July, what would have been its central angle in the pie chart?

A 64.8°

B 72°

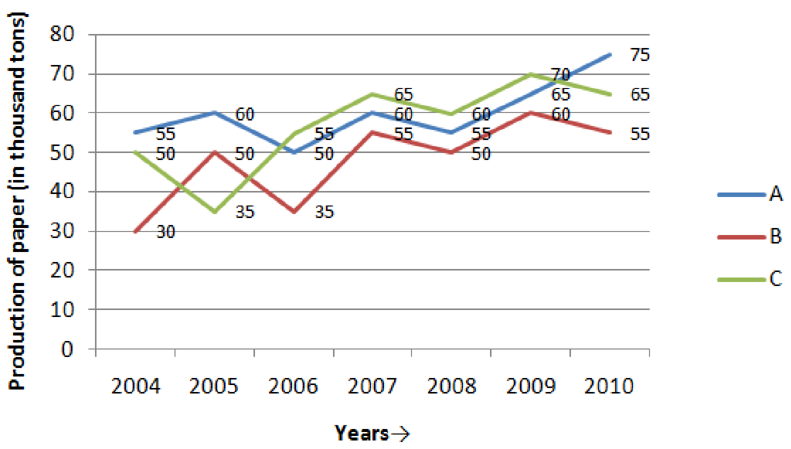
C 60°

D 42.5°

Instructions [31 - 35]

Study the following graph and answer the questions.

Productions of paper (in thousand tons) by three different factories over the years 2004 to 2010



31. Total production of paper of the three factories together is equal for the years

A 2004, 2006

B 2005, 2007

C 2007, 2008

D 2009, 2010

32. In which of the following years for the factory B, the percentage rise or fall in the production from the previous year is maximum?

A 2004

B 2006

C 2007

D 2008

33. Average production of paper for year for the factory A is the what percentage of Avearage production per year for factory c?

A 125

B 119

C 105

D 80

34. Ratio of total production of paper to the factory A to the total production by factory c is

A 20:21

B 21:20

C 80:67

D 84:67

35. In how many years the production of paper (in thousand tons)by factory c is less than the average production of paper by factory B over the years

A 1

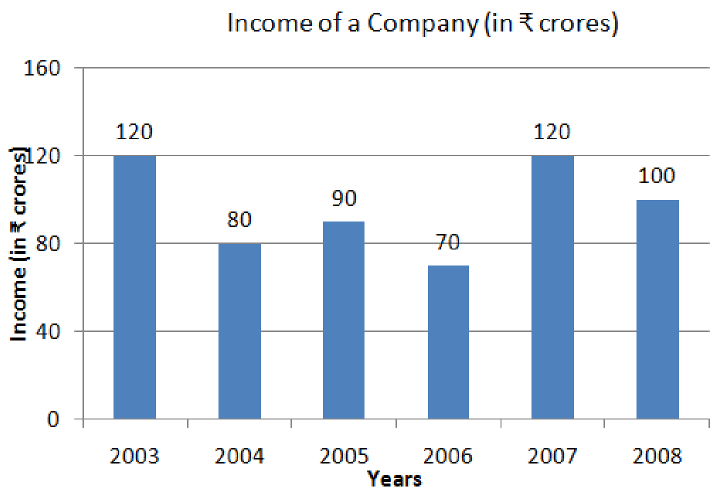
B 2

C 3

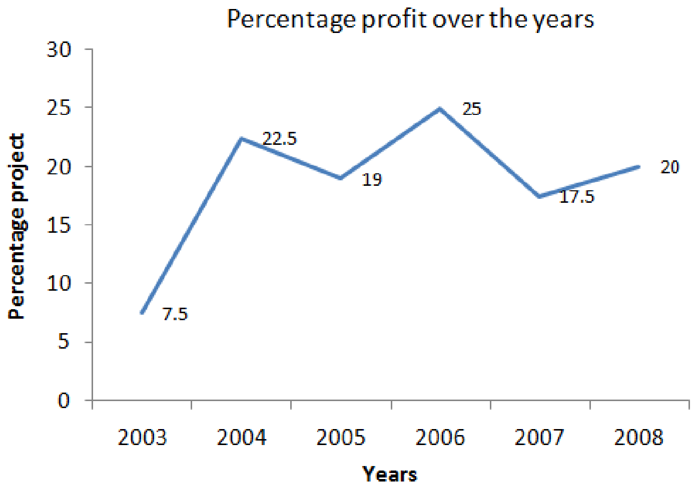
D 4

Instructions [36 - 40]

Study the following graph (i) and (ii) which shows income in(₹ in crores) and percentage profit over the years 2003 to 2008, and answer the questions



(i)



(ii)

36. In which year was the amount of profit was maximum?

A 2004

B 2006

C 2007

D 2008

37. Difference in the amount of profit for 2005 and 2008 is

A 1.67 cores

B 1.87 crores

C 1 core

D 0.31 crore

38. Approcimately what was the average expenditure(in ₹ crores) for the given years?

A 81.9

B 82.2

C 96.7

D 98.7

39. The expenditure (in ₹ crores) in 2006 is

A 83.3

B 75

C 65.31

D 56

40. If the profit percent in 2008 was 25, what would have been the expenditure (in ₹ crores) in that year ?

A 75

B 75.8

C 80

D 83.3

Verbal Ability

Instructions [41 - 44]

In each of the following sentences four words or phrases have been underlined. One underlined part in each sentenceis not acceptable in standard English. Pick up that part and mark its number.

41. The government **initiated/(1)** various **measure/(2)** to **raise/(3)** the living **standards/(4)** of the people.

A (1)

B (2)

C (3)

D (4)

42. The book **must be/(1)** old **for/(2)** its cover **is torn/(3)** **bad/(4)**

A (1)

B (2)

C (3)

D (4)

43. The region has a climate **so severe that/(1)** the plants **growing there/(2)** **had been/(3)** rarely more than twelve inches **high/(4)**

A (1)

B (2)

C (3)

D (4)

44. The **reason/(1)** for my **prolonged/(2)** **absence/(3)** from the class was **because/(4)** I was ill.

A (1)

B (2)

C (3)

D (4)

Instructions [45 - 48]

Fill in the blanks with the most appropriate words.

45. When such .... remarks are circulated, we can only blame and despise those who - produce them.

A adulatory

B chance

C rhetorical

D redundant

46. It would be difficult for one so .... to be led to believe that all men are equal and that we must disregard race, colour and creed.

A emotional

B democratic

C intolerant

D obsolete

47. The enemy soldiers were hot in pursuit; desperate. The fugitive sought .... in the village church.

A salvation

B sanctuary

C therapy

D mercy

48. Fitness experts claim that jogging is .... once you begin to jog regularly, you may be unable to stop, because you are sure to love it more and more all the time.

A exhausting

B illusive

C addictive

D exotic

Instructions [49 - 52]

Each question below consists of a related pair of words followed by four pairs of words.
Select the pair that best expressesthe relatioship similar to the original pair.

49. Clay: mould::

A Wood: carve

B Paper: burn

C Pipe: blow

D Trees: sway

50. Play: Acts::

A Essay: Topics

B Novel: Chapters

C Game: Players,

D Poem: rhymes

51. Clock: time::

A Watch: wrist

B Radio : sound

C Odometer : speed

D Yardstick : distance

52. Gazelle : swift::

A Horse : slow

B Lion : roar

C Lamb: bleat

D Swan : graceful

Instructions [53 - 56]

In the following questions choose the alternative which best expresses the meaning of the given word.

53. Ecstasy

A joy

B treasure

C warmth

D lack

54. Foster

A fondle

B rear

C roll

D speed

55. Credible

A worthy

B believable

C noticeable

D careful

56. Insipid

A lucid

B witty

C why

D flat

Instructions [57 - 60]

Choose the word which is opposite in meaning to the given word.

57. Hubbub

A capital

B fury

C calm

D axle

58. Undulating

A ups

B flat

C steep

D gradual

59. Curtail

A burden

B lengthen

C encourage

D express

60. Resilient

A unyielding

B pungent

C unsavoury

D foolish

Instructions [61 - 64]

In the following questions the word at the top is used in four different ways. Choose the option in which the usage of the word is incorrect or inappropriate.

61. PUSH

A The manufacturers are really pushing this new shampoo.

B He has difficulty pushing his feelings into words.

C We should be able to move this table if we push it together.

D She pushed through the crowd saying that she was a doctor.

62. HARBOUR

A large ship is anchoring in the harbour today.

B The crowing of the cock harbours the dawn.

C Antonio harboured the thoughts of revenge for his brother for doing him wrong.

D Dogs harbour fleas in their thick fur.

63. RIGHT

A What a rightly idea-he had to visit this place!

B You were quite right to refuse this assignment.

C The driver quickly righted the car after it skidded.

D I will come right away, without any delay.

64. MASTER

A She could not master the courage to tell her friend- about her loss

B She quickly mastered the art of interviewing people.

C The terrorist was a master of disguise.

D He is the master of his house.

Instructions [65 - 68]

Fill in the blanks with appropriate alternative.

65. Anurag asked me ....

A why didn't you pay the driver ?

B why I didn't pay the driver.

C why didn't I pay the driver ?

D why you didn't pay the driver.

66. During the summer months,there is

A prolonged country shortage in many parts of power.

B prolonged parts of the country in many power shortage.

C many parts of the country in prolonged power shortage.

D prolonged powershortage in many parts of the country

67. We heard an incessant noise from our neighbours' house. They .... all the time.

A are quarrelling

B were quarrelling

C will be quarrelling

D would be quarrelling

68. The breakdown was said .... a defective transformer.

A to be caused by

B it was caused by

C that the cause was

D to have been caused by

Instructions [69 - 71]

In the following questions, a sentenceis given in active voice. Find the correct passive voice version from the given alternatives.

69. Somebody has put out the light.

A The light is put out.

B The light was being put out

C The light has been put out.

D The light had been put out.

70. Wejustly rebuked him for acting so selfishly.

A He has justly rebuked for selfishness.

B He was justly rebuked by us for acting so selfishly.

C He has been justly rebuked by us for acting so selfishly.

D He has rebuked by us for acting so selfishly.

71. The rules forbid passengers to cross the railway line.

A Passengers are forbidden by the rules to cross the railway line.

B Passengers were forbidden by the rules to cross the railway line.

C Passengers have been forbidden by the rules to cross the railway line.

D Passengers had been forbidden by the rules to cross the railway line.

Instructions [72 - 75]

In the following sentences replace the underlined words with the appropriate expression from the given alternatives.

72. His friends expected that he would *escape* with a fine.

A get off

B go out

C get out

D give way

73. How did these things *happen* ?

A came out

B came about

C came in

D came up

74. Nobody was at home when the fire *started.*

A took off

B came about

C kicked out

D broke out

75. The crew of the boat *underwent* terrible suffering.

A passed through

B went by

C stood up

D pulled together

Instructions [76 - 80]

In the following passage there are blanks which have been numbered. For each blank, four words are suggested. Find out the most appropriate word to fill in the blank.

If you ask some people, "How did you learn English so well ?" You may get a surprising answer : "In my sleep!"

These are ......76 who have taken part in one of the ....77 exeriments to test learn- while-you-sleep methods, ...78 are now being tried out in several countries, and with ....79 subject, of which English is only one. Specialists say that ....80. sleep study method speeds language learning tremendously.

76. 76

A men

B women

C people

D person

77. 77

A recent

B late

C modern

D new

78. 78

A they

B who

C these

D which

79. 79

A all

B several

C both

D different

80. 80

A this

B a

C any

D every

Logical Ability

Instructions [81 - 85]

In questions given below, Assumption ‘A’ is followed by Reasons RI and RII. Apply RI and RII to A and mark your answer as under:

81. A.Since time immemorial more people have been killed or made to suffer in the nameof religion than for any other reasons.
RI.The essenceofreligion is hatred in other religions.
RII.Those who lose their lives in the name of religion go straight to Heaven, so devotees do not mind losing their lives

A Only RI is the reason for A

B Only RII is the reasons for A.

C Both RI and RII are the reasons for A.

D Neither RI nor RII is the reason for A.

82. A. Most of the columms of modern magazines are covered with news of political events.
RI.Readers are very much interested in political turmoil and scandals.
RII.It is not difficult to write about politics as one does not need originality.

A Only RI is the reason for A

B Only RII is the reasons for A.

C Both RI and RII are the reasons for A.

D Neither RI nor RII is the reason for A.

83. A.only Secularism in thought word and deed can save India from Chaos.
RI.Communal differences would only result in nots and destruction of life, faith, property
RII.our democracy Demands open mind and heart for moral, Spiritual and physical uplift of the country

A Only RI is the reason for A

B Only RII is the reasons for A.

C Both RI and RII are the reasons for A.

D Neither RI nor RII is the reason for A.

84. A Vice-Chancellors do not have a very comfortable time.
RI.Students are always causing disciplinary problems.
RII. When people become vice-chancelors they are very old.

A Only RI is the reason for A

B Only RII is the reasons for A.

C Both RI and RII are the reasons for A.

D Neither RI nor RII is the reason for A.

85. A. TV does not give much coverage to activities undertaken by ex-Presidents and Prime Ministers.
RI. Everyone salutes the chair.
RII. Those persons do not perform any activity worth mentioning.

A Only RI is the reason for A

B Only RII is the reasons for A.

C Both RI and RII are the reasons for A.

D Neither RI nor RII is the reason for A.

Instructions [86 - 90]

Find the old-man out

A Idle

B Lethargic

C Lazy

D Subdued

A Blackmail

B Forgery

C Snobbery

D Sabotage

A Milk

B Orange

C Cotton

D Snow

A Cement

B Brick

C Wall

D Stone

A Raw

B Tasty

C Ripe

D Rotten

Instructions [86 - 89]

Match the following questions with (1), (2), (3) and (4) on the basis of similar relationship. The items may not be in the same order.

86. Car : road : petrol

A Post-office : hospital : city

B Army : defence : navy

C Pen : ink ; paper

D Life : alive : dead

87. Day: light : night

A Post-office : hospital : city

B Army : defence : navy

C Pen : ink ; paper

D Life : alive : dead

88. Music : dance : art

A Post-office : hospital : city

B Army : defence : navy

C Pen : ink ; paper

D Life : alive : dead

89. Body : livere : heart

A Post-office : hospital : city

B Army : defence : navy

C Pen : ink ; paper

D Life : alive : dead

Instructions [90 - 94]

A word arrangement machine when given an input line of words, rarranges them following a particular rule in each step. The following is the illustration of the input and the steps of arragement:
Input : She was shot dead at, her residence

Step I : at was shot dead she her residence
Step II : at her shot dead she was residence
Step III : at her she dead shot was residence
Step IV : at her she was shot dead residence
Step V: at her she was shot dead residence
Since the words are fully arranged, the machine stops. Otherwise it may go on till the words get fully arranged.
Study the logic and answer the questions.

90. In how many steps will the following input be fully arranged ?
Input : India has always been a critical factor

A Three

B Four

C Five

D Six

91. What would be the Step III for the following input ?
Input : this is one thing on which I caution

A I is one thing on which this caution

B I is on one this which thing caution

C I is on thing one which this caution

D I is one thing which this caution

92. If step II of an input reads "by he out the efforts made spells us," which of the following would be the last step ?

A step III

B step IV

C step V

D step VI

93. What could be the penultimate step for the following input ?
Input : You hardly see any motorised vehicle

A step I

B step III

C step IV

D step VI

94. What could be the step IV for the following input ?
Input : the foliage along road can deceive you

A can foliage along road the deceive you

B can the you road along the deceive foliage

C can the you road foliage deceive along

D can the along road foliage deceive you

Instructions [95 - 100]

Study the following information and answer the questions given below?
In a certain code, the symbol for 0 (Zero) is @ and for 1 is $. There are no other symbols for all other numbers greater than one. The numbers greater then | are to be written only by using the two symbols given above. The value of the symbol for 1 doubles itself every time it shift one place to the left. Study the following examples:
'0' is written as @
'1' is written as $
'2' is written as $ @
'3' is written as $ $
'4' is written as $ @@ and so on

95. Which of the following numbers will be represented by $ $ @ $?

A 8

B 11

C 13

D 12

96. Which of the following will represent the value of 3 x 3 +1 ?

A $ $ @ $

B $ @ $ @

C $ @ @ $ $

D $ $ $

97. Which of the following numbers will be represented by $ @ @ @ $?

A 22

B 31

C 14

D 17

98. What is the average of \$\$\$@@ and \$\$@@

A \$@@@@

B @

C \$@\$@@

D \$@@\$@

99. What is the value of $[(\$@) + (\$@) \div (\$@)]^{\$\,@}$?

A $\$@@\$@$

B $\$@\$@\$$

C $\$\$\$ \times \$\$ \quad - \$@$

D $(\$\$)^{\$@}$

100. What is the value of $[(8 + 16) \div (4 \times \ 3)]^{3}$?

A $\$@@\$$

B $\$\$@@$

C $\$\$\$@$

D $\$@@@$

Instructions [101 - 105]

In each question below, there are three statements followed by four conclusions numbered I, II, III and IV. You have to take the given statements to be true even if they seem to be at variance with commonly known facts and then decide which of the given conclusions logically follow (s) from the given statement.

101. Statements : Some charts are darts.
All darts are carts.
Some carts are smarts.
Conclusions : I. Some charts are carts.
II. Some carts are darts -
III. Some darts are smarts.
IV. Some smarts are charts.

A Only I and III follow

B Only II and III follow

C I and II follow

D I, III and IV follow

102. Statements : All boxes are tables.
No desks are tables.
Some desks are curtains.
Conclusions : I. No Boxes are desks.
II. Some Boxes are desks.
III. Some curtains are not boxes
IV. Some curtains are boxes.

A III and either I or II follow

B I and either III and IV follow

C Either I or II and either III and IV follow

D I and III follow

103. Statement: No killer is a sweater.
No jacket is a sweater.
Some jacket are roses.
Conclusions : I. Some roses are sweters
II. some roses are not swaters.
III. No killer is a jucked.
IV. Some jackels are killers.

A Either I or II and III follows

B Either III or IV or IL follow

C Either I and III follow

D Either I or II and either III or IV follow

104. Statements: No student is decent
Some decent are bags.
All bags are roses
Conclusions: I. Some bags are not students.
II. All bags are students.
III. Some decents are roses.
IV. All roses are decent.

A Only i follows

B On Either I or II follows

C Either I or II, and III follows

D I and II follow

105. Statements : Some birds are stones.

Some, tigers are birds.

All stones are grapes.

Conclusions : I. Some stones are birds.

II. Some stones ,are not birds.

III. Some grapes are birds.

IV. Some tigers are stones

A I, I, and II follow

B I, II and IV follow

C Either I or II and III folow

D I and II follow

Instructions [106 - 110]

A researcher studying organic compounds has found that five different molecules. T, W, X, Y and Z form chains according to the following rules :
A chain consists of three or more molecules, though the molecules in the chain are not necessarily different.
T is never found on either end of a chain.

If W appears in a chain, it appears more than once.
X is never found next to y in a chain.
W is never found on the end of a chain
unless Z is found somewhere in the chain.
If Y appears in a chain, Z appears also,

106. Whichof the following is a possible chain of molecules ?

A T X Y Z

B Y T X X

C W Z T Y

D W W X Z

107. Which of the followings is NOT possible chain of molecules ?

A X X T Z

B Z X W W Z

C W X Z Y W

D Y W T Z X X

108. The chain W W T Y Z X can be changed into another chain by carrying out any one of the following EXCEPT

A replacing the T molecule with a molecule

B replacing the Y molecule with an X molecule

C replacing the X molecule with a T molecules

D interchanging the T and the Z molecules

109. Which of the following is not a chain but could be turned into a chain by changing the order of the molecules ?

A X Y T X

B T X X Y

C W T T W

D W X X W

110. Which of the following sequences can be converted into a chain by adding Z and rearranging the molecules ?
I. X Y X T

II. W T T Z

III. X X Y W

A I only

B I and III only

C III only

D I, II and III

Instructions [111 - 115]

In the following questions the symbols @, @, =, © and © are used with the following meaning :

P @ Q means P is greater than Q.

p @ Q means P is either greater than or equal to Q.

P = Q means is equal to Q.

P © Q means P is smaller than Q.

P © Q means P is either smaller than or equal to Q.

For each questions you have to assume given statements to be true and then decide which of the two given conclusions is / are definitely true. Give answer :

111. Statements: B @ V,K © C, C © B
Conclusions:

I. V @ C

II. B @ K

A if only conclusion II is true

B if either conclusion I or II is true

C if neither I nor II is true

D if both the conclusions are true

112. Statements : K @ T, S = K, T © R
Conclusions :

I.S @ R

II. T = R

A if only conclusion II is true

B if either conclusion I or II is true

C if neither I nor II is true

D if both the conclusions are true

113. Statements : U = M, P @ U, M @ B
Conclusion :

I. P = B

II. P @ B

A if only conclusion II is true

B if either conclusion I or II is true

C if neither I nor II is true

D if both the conclusions are true

114. Statements: L @ N, J © P, P @ L
Conclusion:

I. J = L

II. P = N

A if only conclusion II is true

B if either conclusion I or II is true

C if neither I nor II is true

D if both the conclusions are true

115. Statements:H @ G, D @ E, H = E
Conclusion:

I. D @ H

II. G @ D

A if only conclusion II is true

B if either conclusion I or II is true

C if neither I nor II is true

D if both the conclusions are true

Quantitative Ability

116. The number 123456789 and 999999999 are multiplied. How many of the digits in the product are 9's

A 0

B 1

C 2

D 3

117. If the expression $15^6 \times 28^5 \times 55^7$ is evaluated, the number of zeros at the end of the number is

A 8

B 10

C 18

D 20

118. $\sqrt[4]{2001\ 2001\ 2001\ 2001}$ is closest to

A 2001

B 2100

C 6700

D 10010

119. The units digit of $(2002)^{2002}$ is

A 2

B 4

C 6

D 8

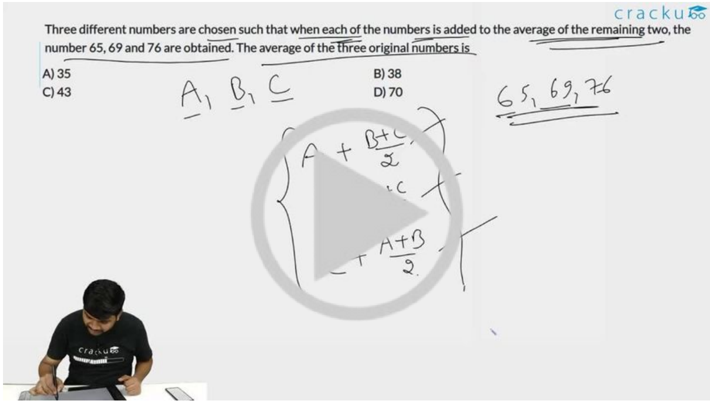
120. Three different numbers are chosen such that when each of the numbers is added to the average of the remaining two, the number 65, 69 and 76 are obtained. The average of the three original numbers is

A 35

B 38

C 43

D 70



VIDEO SOLUTION

121. *abc* and *def* are 3-digit numbers such that

$$\begin{array}{ccc} a & b & c \\ d & e & f \\ \hline \multicolumn{1}{|c|}{10} & \multicolumn{1}{c|}{0} & \multicolumn{1}{c|}{0} \\ \hline \end{array}$$

and none of *a, b, c, d, e,* or *f* is 0. What is the sum $a + b + c + d + e + f$?

A 10

B 19

C 21

D 28

122. When a number is divided by 5, the remainder is 2, when divided by 7, the remainder is 3, when divided by 9, the remainder is 4. The sum of digits of such smallest number is

A 13

B 11

C 9

D 7

123. The expression

$\frac{1}{8}+\frac{1}{10}+\frac{1}{11}+\frac{1}{15}+\frac{1}{20}+\frac{1}{41}+\frac{1}{110}+\frac{1}{1640}$ is equal to

A 1

B $\frac{13}{15}$

C $\frac{11}{15}$

D $\frac{7}{15}$

124. Let r be the least non-negative remainder when $(22)^7$ is divided by 123. The value of r is

A 5

B 22

C 32

D 52

125. if $\frac{97}{19}=w+\frac{1}{x+\frac{1}{y}}$ where x, y and w are all positive integers, the value of $x+2y-3w$ is

A 2

B -2

C 3

D -5

126. Let $a=2^{129}\times 3^{81}\times 5^{128}$, $b=2^{127}\times 3^{81}\times 5^{128}$, $c=2^{126}\times 3^{82}\times 5^{128}$, and $d=2^{125}\times 3^{82}\times 5^{129}$. Then

A $b>c>d>a$

B $b<c<d<a$

C $b>c>a>d$

D $b<c<a<d$

127. The positive integer a is a 2-digit number (01, 02 are not 2-digit number) the positive integer b has 'a' digit and the positive integer 'c' has 'b' digits. The smallest possible value for c is

A 10^{9^9}

B 10^{10^9}

C $10^{10^{10}} - 1$

D $10^{10^9 - 1}$

128. If $2^{36} - 1 = 68a$ 19476735, when all the digits are correct except a, the correct value of a is

A 9

B 8

C 7

D 3

129. The value of

$\frac{(1.121)^3 - (3.333)^3 + (2.212)^3}{(1.121)(3.333)(2.212)}$ is

A 1

B -2

C -3

D 4

130. If $a : b = 3 : 4, b : c = \frac{8}{9}$ and $c : d = \frac{2}{3}$, then the value of $\sqrt[4]{\frac{ad}{b^2}}$ is

A $\frac{3}{4}$

B $\frac{9}{8}$

C $\frac{3\sqrt{3}}{4}$

D $\frac{3\sqrt{2}}{4}$

131. The value of $\frac{\sqrt{43 - 12\sqrt{7}} - 2}{16 + 6\sqrt{7}}$ is

A 1

B 2

C 3

D $3 - 2\sqrt{7}$

132. Let N be the greatest natural number that will divide 13511, 13903 and 14589 leaving same remainder in each case. The sum of digits of N is

A 10

B 13

C 15

D 17

133. The value of
$\frac{(0.251)^2-(0.051)^2-(0.511)^2-2(0.051)(0.511)}{(0.251)^2-(0.051)^2-2(0.251)(0.051)-(0.511)^2}$ is

A $\frac{271}{237}$

B $\frac{237}{271}$

C $\frac{823}{711}$

D $\frac{701}{823}$

134. The cost of making a rectangular table is calculated by adding two variables. The first is proportional to the area of the table and the other to the square of the length of the longer side. In making 2 m x 3 m table it costs ₹ 5,000 and in making a 1.5 m x 4 m table, it cost ₹ 6,400. The cost of making a 2.5 m x 2.5 m table is (nearest to a rupee)

A ₹ 3,383

B ₹ 4,583

C ₹ 4,853

D ₹ 4,835

135. Anu is walking at a constant speed halfway between two paralled train tracks. On each track is a train of the same length. They are approaching Anu from different directions both at the same speed v km/hour. The train going in the same direction as Anu going takes t_1, second to pass her, while the other takes t_2, seconds to pass her. Speed of Anu (in km/hour) is

A $\frac{t_1-t_2}{t_1+t_2}v$

B $\frac{t_2-t_1}{t_1+t_2}v$

C $\frac{t_1+t_2}{t_1-t_2}v$

D $\frac{t_1+t_2}{t_2=t_1}v$

136. A multiple choice examination consists of 20 questions. The scoring is +5 for each correct answer, —2 for each incorrect answer, and 0 for each unanswered question. A student's score on the examination is 48. The maximum number of questions he could have answered correctly is

A 14

B 12

C 10

D 9

137. At an institute, 99% of the 100 students are girls but only 98% of the students living on the compus are girls. If same girls live on campus, how many students live off compus ?

A 2

B 40

C 50

D 98

138. A and B share a piece of land. The ratio of the area of A's portion to the area of B's portion is 2 : 3. They each grow wheat and barley on their pieces of land. The entire land is covered by wheat and barley in the ratio 7 : 3. On A's land, the ratio of wheat do barley is 4: 1. The raito of wheat to barley for B's land is

A 11:19

B 19:11

C 11:9

D 9:11

139. A box has apples and oranges. $\frac{2}{3}$ of all the apples and $\frac{3}{4}$ of all the oranges are rotten The number of rotton apples equals the number of rotton oranges. What fraction of the total number of fruits in the box is rotton ?

A $\frac{12}{17}$

B $\frac{5}{17}$

C $\frac{9}{13}$

D $\frac{11}{13}$

140. A TV set is available for ₹ 19,650 cash payment or for ₹ 3,100 cash down payment and three equal annual instalments. If the interest is charged at the rate of 10% per annum compounded annually, the amount of each instalment is

A ₹ 6,555

B ₹ 6,612

C ₹ 6,655

D ₹ 6,665

141. A person bought n articles for ₹ d. He sold two articles at half their cost and the rest at a profit of ₹ 8 on each article. If the overall profit is ₹ 72, there the least possible value of n is

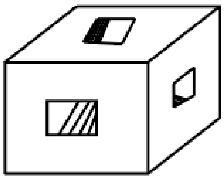
A 12

B 10

C 8

D 16

142. A 3 cm x 3 cm x 3 cm cube has three holes each of 1 cm x 1 cm cross-section running from the centre of each face to the centre of the opposite face. The total surface area (in cm^2) of the solid so obtained is



A 48

B 50

C 72

D 78

143. If a 6 cm X 6 cm square is placed on a triangle, it can cover up to 60% of the triangle. If the triangle is placed on the square it can cover up to $\frac{2}{3}$ of the square. The area (in cm^2) of the triangle is

A 24

B 36

C 40

D 60

144. A circular grass plot 4 m in diameter is cut by a straight path | m wide, one edge of which passes through the centre of the plot. Area of the remaining portion is

A $\frac{4\pi}{3} - \sqrt{3}$

B $\frac{10\pi}{3} - \sqrt{2}$

C $\frac{10\pi}{3} + \sqrt{3}$

D $\frac{10\pi}{3} - \sqrt{3}$

145. Three concentric circles have radii (in cm) a, b and c, where a < b < c. If a = 8 and b = 9 and the middle circle bisects the area between the other two circles, then the value of c is

A $7\sqrt{2}$

B $6\sqrt{3}$

C $7\sqrt{3}$

D 10

146. A sphere has a diameter of $500\sqrt{3}$ cm. A biggest cube is fitted in it. Now a biggest sphere is fitted within this cube. Again a biggest cube is fitted in the smaller sphere. The ratio of the valume of bigger cube to the valume of smaller cube is

A $3:1$

B $2\sqrt{3}:1$

C $3\sqrt{3}:1$

D $4\sqrt{3}:1$

147. A rectangular box has dimensions x, y and z units, where x < y < z. If one dimension only is increased by one unit, then the increase in volume is

A greatest when x is increased

B greatest wheny y is increased

C greatest when z is increased

D the same regardless of which dimension is increased.

Instructions [148 - 151]

Read the following table and answer questions.

School	Number of Students scoring less than 60% marks	Percentage of Students scoring more than 60% marks	Total number of Students appeared
A	320	55	800
B	220	40	400
C	300	20	375
D	280	10	350
E	210	25	300

148. Which school has the lowest percentage of students scoring less than 60% marks ?

A A

B B

C D

D E

149. Number of schools which have the same percentage of students scoring exactly 60% is

A 1

B 2

C 3

D 4

150. Total number of students scoring 60% or more marksis

A 110

B 775

C 785

D 895

151. The ratio of the total number of students scoring less than 60% marks to that of scoring exactly 60% marks is

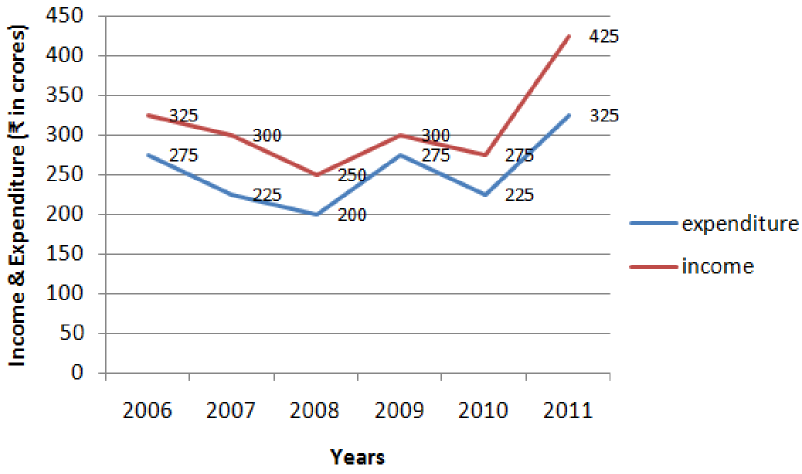
A 11:133

B 133:11

C 157:22

D 285 : 179

Instructions [152 - 155]



The above graph shows income and expenditure (₹ in crores) of a company in the years 2006 to 2011. Read the graph and answer the questions

152. The total expenditure of which of the following pairs of years was equal to the income in 2011 ?

A 2006 and 2007

B 2007 and 2008

C 2006 and 2008

D 2007 and 2010

153. In how many of the given years was the income more than the average income of the given years ?

A 1

B 2

C 3

D 4

154. What was the approximate percentage increase expenditure from 2010 to 2011 ?

A 72.3

B 70.5

C 54.5

D 44.4

155. In which year the percentage of expenditure to income, the highest?

A 2009

B 2008

C 2007

D 2006

Answers

Data Interpretation

1.**A**	2.**B**	3.**D**	4.**C**	5.**D**	6.**C**	7.**C**	8.**B**
9.**A**	10.**D**	11.**A**	12.**C**	13.**D**	14.**B**	15.**D**	16.**C**
17.**B**	18.**A**	19.**D**	20.**A**	21.**B**	22.**C**	23.**C**	24.**A**
25.**D**	26.**A**	27.**C**	28.**D**	29.**B**	30.**A**	31.**D**	32.**A**
33.**C**	34.**B**	35.**A**	36.**C**	37.**A**	38.**B**	39.**D**	40.**C**

Verbal Ability

41.**B**	42.**D**	43.**C**	44.**D**	45.**B**	46.**C**	47.**B**	48.**C**
49.**A**	50.**B**	51.**D**	52.**D**	53.**A**	54.**B**	55.**B**	56.**D**
57.**C**	58.**B**	59.**B**	60.**A**	61.**B**	62.**B**	63.**A**	64.**A**
65.**B**	66.**D**	67.**B**	68.**A**	69.**C**	70.**B**	71.**A**	72.**A**
73.**B**	74.**D**	75.**A**	76.**C**	77.**A**	78.**D**	79.**B**	80.**A**

Logical Ability

81.**D**	82.**A**	83.**D**	84.**A**	85.**B**	86.**D**	87.**C**	88.**A**
89.**C**	90.**B**	91.**C**	92.**D**	93.**B**	94.**A**	95.**B**	96.**B**
97.**C**	98.**B**	99.**B**	100.**C**	101.**B**	102.**D**	103.**C**	104.**D**
105.**D**	106.**C**	107.**D**	108.**D**	109.**D**	110.**D**	111.**D**	112.**D**
113.**C**	114.**D**	115.**A**	116.**A**	117.**C**	118.**B**	119.**C**	120.**D**

Quantitative Ability

121.**A**	122.**B**	123.**C**	124.**B**	125.**A**	126.**D**	127.**A**	128.**D**
129.**D**	130.**B**	131.**B**	132.**D**	133.**C**	134.**C**	135.**D**	136.**C**
137.**D**	138.**A**	139.**B**	140.**A**	141.**B**	142.**C**	143.**B**	144.**A**
145.**C**	146.**A**	147.**C**	148.**C**	149.**D**	150.**A**	151.**C**	152.**A**
153.**A**	154.**C**	155.**D**	156.**B**	157.**B**	158.**B**	159.**D**	160.**C**

Explanations

Data Interpretation

Explanation [1 - 5]:
Let's tabulate all the information from the given table:

The calculation is as follows:

Consider the male 18 - 25 range:

=> Total consumers = 600

=> Out of them, 240 like A, 180 like B, 90 like both

=> Number of people who like either = 240 + 180 - 90 = 330

=> 600 - 330 = 270 people don't like any of the drinks.

Doing the same calculation for all the categories and tabulating, we get:

Male	Total Surveyed	A	B	Both	Either	None
18 - 25	600	240	180	90	330	270
26 - 40	840	360	480	90	750	90
41 - 60	1200	300	240	120	420	780

Female	Total Surveyed	A	B	Both	Either	None
18 - 25	720	180	120	60	240	480
26 - 40	540	120	360	180	300	240
41 - 60	960	240	300	180	360	600

1. **A**
In the age group 26 - 40, Males: 840 - 480 = 360 did not prefer B, Females: 540 - 360 = 180 did not prefer B, in total 360 + 180 = 540 is the required answer.

2. **B**
X.png"/>
In the 41-60 category: The number of consumers who preferred either of the two drinks = 420 + 360 = 780.

3. **D**
X.png"/>
In 18-25:

Males: Number of people who only prefer 1 drink = 240 + 180 - 2*90 [subtracting those who like both] = 240

Females: 180 + 120 - 2*60 = 180

=> Required answer = 240 + 180 = 420.

4. **C**
X.png"/>
Total consumers above 40 years old = 1200 + 960 = 2160; out of these, 780 + 600 = 1380 preferred none of the drinks.

=> Required %age = 1380/2160 * 100 = 63.88 = 69%

5. **D**
X.png"/>
Total consumers above 40 years old = 1200 + 960 = 2160; out of these, 780 + 600 = 1380 preferred none of the drinks.

=> Total number of consumers who did not prefer any drink is 270 + 90 + 780 + 480 + 240 + 600 = 2460

Verbal Ability

Logical Ability

81. **D**
The assertion talks about people being killed in the name of religion, among other reasons. However, it doesn't state anything about hatred or that people who are dying shall straightaway go to heaven. Hence, none of the given reasons are correct.

82. **A**
The assertion talks about the abundance of columns on political events. Hence, it is clear that the readers are in the same, else this would not have been the case. However, we cannot comment on the originality, as those columns also require a certain level of effort/originality. Hence, only R1 is correct.

83. **D**
The assertion talks about the need for secularism in India. R1 is too narrow by the use of 'only', while R2 is vague as we cannot say anything about the requirements of democracy and having an open mind/heart. R2 is totally disconnected with the idea of secularism.

84. **A**
The assertion talks about Vice-chancellors not having a very comfortable lives.

R1 can be a possible reason since the VC is supposed to handle the internal affairs of a college. However, R2 is incorrect because the assertion does not talk about discomfort in sense of being old.

85. **B**
The assertion talks about lesser coverage of the ex-president/PM.

R1 is incorrect, as saluting the chair shall not be the reason for not covering ex-members if they are performing significant functions.

R2 is correct, as the ex-members would not have many noteworthy activities under their portfolio.

86. **D**
Lazy, lethargic, and idle are synonyms. However, subdued means calm, which is entirely different in terms of the connotation.

87. **C**
Sabotage, blackmail, and forgery are criminal conduct. On the other hand, snobbery is an attitude in which one likes people of higher social status.

88. **A**
Orange, cotton, and snow are found naturally in the environment. However, milk is extracted from lactating animals through special efforts.

91. **C**
As car operates on petrol, and runs on road: Pen operates on ink and writes on a paper.

92. **D**
As day signifies light, and the opposite is night: Life signifies being alive, and the opposite is dead.

93. **B**
As music and dance are art forms, the Army and navy are a part of the defense.

94. **A**
Just like liver and heart are situated in the body, post office and hospital are situated in the city.

106. **C**
We shall test each of the conclusions:

1. Since all darts are carts and some darts are charts, it is definitely true that the charts that are darts shall be carts as well.

2. Since all darts are carts, this statement is true.

3. It is possible that the carts which are smart might not be darts at all.

4. Same logic as 3.

107. **D**
1. Since all boxes are tables and no table is a desk, it is clear that no box shall be a desk. Hence, true.

2. Invalid with same logic as 1.

3. We know that no table is a desk, making no box a desk. Hence, the curtains which are desks cannot be boxes.

4. We can conclude 3 but we don't have much information to say whether some curtains are boxes.

116. **A**
B>V, K<C, C<B

1. We know that both V and C are less than B. But we don't have any specific relation between V and C.

2. We know, B>C>K, hence B>K. Hence, true.

117. **C**
K>T, S=K, T<R or T=R

1. We can conclude S>T. However, both S and R can be greater than T (or R=T), and hence, we cannot say that this conclusion is true.

2. This, again, is not certain.

118. **B**
U=M, P>U or P=U, M>B or M=B

from the given information, there are two possibilities as for both P/U and M/B, there are two cases.

Either P=B or P>B.

119. **C**
L>N or L=N, J<P or J=P, P>L or P=L

1. Since there are two cases of P and L, we cannot say that J=L.

2. Same logic as 1.

120. **D**
H>G or H=G, D>E, H=E

1. Since D>E=H, hence D>H.

2. G>E or G=E, and D>E, we can say that D>G.

Quantitative Ability

121. **A**

To find : $123456789 \times 999999999$

= $123456789 \times (1000000000 - 1)$

= $123456789000000000 - 123456789$

= 123456788876543211

Thus, there are **0** 9's

=> Ans - (A)

122. **B**

Expression : $15^6 \times 28^5 \times 55^7$

Number of 2's in the expression (in 28^5) = $(2)^{10} = 10$

Number of 5's = $(5)^6 \times (5)^7 = 13$

$\because$ number of 2's are less than number of 5's, thus number of zeroes will depend on two, hence number of zeroes are **10**

=> Ans - (B)

123. **C**

Since options are not so close, we can think of the 4th root of $2 \times \ 10^{15}$

This is nothing but 4th root of 2000 * 10^{12}

=> 1000 * 4th root of 2000 [between 6 and 7] => Answer lies between 6000 and 7000 => Option-C

124. **B**

Expression : $(2002)^{2002}$

It can be written as : $(2002)^{4n+2}$

Thus, the unit's digit will be same as $(2)^2 = 4$

=> Ans - (B)

125. **A**

Let the three original numbers be x, y, z

According to ques, => $x + (\frac{y+z}{2}) = 65$

Similarly, $y + (\frac{x+z}{2}) = 69$

and $z + (\frac{y+x}{2}) = 76$

Adding all the equations, => $(x + y + z) + (x + y + z) = 65 + 69 + 76$

=> $x + y + z = \frac{210}{2} = 105$

Dividing both sides by 3, => $\frac{x+y+z}{3} = \frac{105}{3} = 35$

$\therefore$ Required average = **35**

=> Ans - (A)

VIDEO SOLUTION

126. **D**

Here, $c+f=10$ and $b+e=9$ and $a+d=9$ [because of carry over 1]

Adding all the equations, => $a+b+c+d+e+f=10+9+9=28$

=> Ans - (D)

127. **A**

A number is divided by 9 if sum of its digits is 9

When a number is divided by 9 and leaves no remainder, => sum of digits is 9.

So, when a number is divided by 9 and leaves remainder 4 , then sum of its digits is $9+4=13$

=> Ans - (A)

128. **D**

Expression : $\frac{1}{8}+\frac{1}{10}+\frac{1}{11}+\frac{1}{15}+\frac{1}{20}+\frac{1}{41}+\frac{1}{110}+\frac{1}{1640}$

= $(\frac{1}{41}+\frac{1}{1640})+(\frac{1}{10}+\frac{1}{20})+(\frac{1}{11}+\frac{1}{110})+(\frac{1}{8}+\frac{1}{15})$

= $(\frac{40+1}{1640})+(\frac{2+1}{20})+(\frac{10+1}{110})+(\frac{15+8}{120})$

= $\frac{1}{40}+\frac{3}{20}+\frac{1}{10}+\frac{23}{120}$

= $\frac{3+18+12+23}{120}=\frac{56}{120}=\frac{7}{15}$

=> Ans - (D)

129. **D**

When $(22)^7$ is divided by 123

= $\frac{(22)^2\times(22)^2\times(22)^2\times22}{123}$

Now, when we divide $(22)^2=484$ by 123, remainder is -8

= $\frac{-8\times-8\times-8\times22}{123}$

= $\frac{-20\times22}{123}$

=> Remainder = -(-52) = **52**

=> Ans - (D)

130. **B**

Expression : $\frac{97}{19}=w+\frac{1}{x+\frac{1}{y}}$

Breaking the L.H.S. expression = $5+\frac{2}{19}$

= $5+\frac{1}{\frac{19}{2}}$

= $5+\frac{1}{9+\frac{1}{2}}$

Comparing above equation with : $w+\frac{1}{x+\frac{1}{y}}$

=> $w=5,x=9,y=2$

To find : $x+2y-3w$

= $9+2(2)-3(5)=9+4-15=-2$

=> Ans - (B)

131. **B**

$a = 2^{129} \times 3^{81} \times 5^{128}$

$b = 2^{127} \times 3^{81} \times 5^{128}$

$c = 2^{126} \times 3^{82} \times 5^{128}$

$d = 2^{125} \times 3^{82} \times 5^{129}$

Eliminating common factors $(2)^{125} \times (3)^{81} \times (5)^{128}$

=> $a = 16, b = 4, c = 6, d = 15$

$\therefore b < c < d < a$

=> Ans - (B)

132. **D**
Smallest number having n digits is $(10)^{n-1}$

Smallest possible value of $a = 10$

Now, b has 10 digits, thus smallest possible value of $b = (10)^9$

Now, c has $(10)^9$ digits, thus smallest possible value of $c = (10)^{10^9-1}$

=> Ans - (D)

133. **C**
Expression : $(2)^{36} - 1$

= $(2^3)^{12} - 1 = (8)^{12} - 1$

Now, to check whether the number is divisible by 9, we have : $(-1)^{12} - 1 = 0$

Thus, $(2)^{36} - 1$ is divisible by 9, hence sum of its digits is also divisible by 9.

=> $6 + 8 + a + 1 + 9 + 4 + 7 + 6 + 7 + 3 + 5 = 56 + a$

Thus, $56 + a = 63$

=> $a = 7$

=> Ans - (C)

134. **C**
Expression : $E = \frac{(1.121)^3 - (3.333)^3 + (2.212)^3}{(1.121)(3.333)(2.212)}$

Let $x = 1.121, y = 2.212, z = -3.333$ and hence $x + y + z = 0$ ------------(i)

=> $E = -\frac{(x)^3 + (y)^3 + (-z)^3}{xy(-z)}$ ------------(ii)

We know that, $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

=> $x^3 + y^3 + z^3 = 3xyz$ -------(iii) $[\because x + y + z = 0]$

Thus, substituting value from equation (iii) in equation (ii), we get : $E = -3$

=> Ans - (C)

135. **D**
Given = $a : b = 3 : 4, b : c = \frac{8}{9}$ and $c : d = \frac{2}{3}$

Multiplying equation (iii) by 9 and equation (ii) by 2, and (i) by 4

=> $a : b : c : d = 12 : 16 : 18 : 27$

To find = $\sqrt[4]{\frac{ad}{b^2}}$

= $\sqrt[4]{\frac{12\times27}{16\times16}}$

= $\sqrt[4]{\frac{81}{64}}$

= $\frac{3}{2\sqrt{2}} = \frac{3\sqrt{2}}{4}$

=> Ans - (D)

136. **C**

Expression : $\frac{\sqrt{43-12\sqrt{7}}-2}{16+6\sqrt{7}}$

= $\frac{\sqrt{(\sqrt{7})^2+(6)^2-2(\sqrt{7})(6)}-2}{(\sqrt{7})^2+(3)^2+2(\sqrt{7})(3)}$

= $\frac{\sqrt{(6-\sqrt{7})^2}-2}{(3+\sqrt{7})^2}$

= $\frac{4-\sqrt{7}}{(3+\sqrt{7})^2}$

137. **D**

N divides each of 13511, 13903, 14589 leaving remainder r in each case.

=> $13511 = aN + r$ -----------(i)

$13903 = bN + r$ --------------(ii)

$14589 = cN + r$ --------------(iii)

Subtracting above equations, we get : $392 = (b-a)N$

$686 = (c-b)N$

$1078 = (c-a)N$

Thus, N must be a factor of all three of these numbers. Thus, $N =$ H.C.F. (392,686,1078) = 98

$\therefore$ sum of digits of N = $9 + 8 = 17$

=> Ans - (D)

138. **A**

Expression : $\frac{(0.251)^2-(0.051)^2-(0.511)^2-2(0.051)(0.511)}{(0.251)^2-(0.051)^2-2(0.251)(0.051)-(0.511)^2}$

Let $x = 0.251, y = 0.051, z = 0.511$

= $\frac{x^2-(y^2+z^2+2yz)}{(x^2-y^2-2xy)-z^2}$

= $\frac{x^2-(y+z)^2}{(x-y)^2-z^2}$

= $\frac{(x+y+z)(x-y-z)}{(x-y-z)(x-y+z)}$

= $\frac{x+y+z}{x-y+z}$

= $\frac{0.251+0.051+0.511}{0.251-0.051+0.511} = \frac{0.813}{0.711} = \frac{271}{237}$

=> Ans - (A)

139. **B**

Let cost of making a table = $C = X + Y$ -----------(i)

Now, $X = k_1 A$, where A is area of table and $Y = k_2 l^2$, where l is length of longer side of table

In making 2 m x 3 m table it costs ₹ 5,000

=> $C = (6k_1) + (9k_2) = 5000$ ------------(ii)

Similarly, $6k_1 + 16k_2 = 6400$ -------------(iii)

Subtracting equation (ii) from equation (iii), we get :=> $7k_2 = 1400$

=> $k_2 = \frac{1400}{7} = 200$

Similarly, $k_1 = \frac{3200}{6}$

Now, for a table having dimension 2.5 m x 2.5 m

=> Cost = $(6.25 \times \frac{3200}{6}) + (6.25 \times 200)$

= $6.25 \times (\frac{3200}{6} + 200)$

= $6.25 \times \frac{4400}{6} \approx Rs. 4,583$

=> Ans - (B)

140. **A**

Let Anu's speed = s km/hr

Let length of each train = l km and speed of each train = v km/hr

Relative speed of train going in same direction = $(v - s)$ km/hr

=> Time taken = $t_1 = \frac{l}{v-s}$ --------------(i)

Similarly, $t_2 = \frac{l}{v+s}$ ------------(ii)

Comparing equations (i) and (ii),

=> $t_1(v - s) = t_2(v + s)$

=> $vt_1 - st_1 = vt_2 + st_2$

=> $s(t_2 + t_1) = v(t_1 - t_2)$

=> $s = \frac{v(t_1 - t_2)}{(t_2 + t_1)}$

=> Ans - (A)

141. **B**

Let questions answered correctly = x and unanswered questions = y, => incorrect questions = $(20 - x - y)$

According to ques,

=> $5x - 2(20 - x - y) = 48$

=> $7x + 2y = 88$

Since, sum of 2 positive numbers is positive, and 84 is the nearest multiple of 7 (when $y = 2$)

=> $7x = 84$ => $x = 12$

Thus, maximum number of questions answered correctly = **12**

=> Ans - (B)

142. **C**

Total students = 100 and Number of girls = 99, Number of boys = 1

Let number of students living off campus = x, and number of students living on campus = $(100 - x)$

2% of boys live on campus (since there is only 1 boy)

=> $\frac{2}{100} \times (100 - x) = 1$

=> $100 - x = 50$

=> $x = 50$

=> Ans - (C)

143. **B**

Let total area of land = 100 sq. units

=> Total area of A's land = 40 sq. units and B's land = 60 sq. units

Also, Total wheat in the land = $\frac{7}{10} \times 100 = 70$ sq. units and similarly, total barley = 30 sq. units

A' land : Wheat = $\frac{4}{5} \times 40 = 32$ sq. units

Barley = $40 - 32 = 8$ sq. units

=> Remaining wheat = $70 - 32 = 38$ sq. units and remaining barley = $30 - 8 = 22$ sq. units

$\therefore$ Ratio of wheat to Barley for B's land = $\frac{38}{22} = 19 : 11$

=> Ans - (B)

144. **A**

Let number of apples = $3x$ and number of oranges = $4y$

=> Number of rotten apples = Number of rotten oranges

=> $2x = 3y$

=> $\frac{x}{y} = \frac{3}{2}$

Let $x = 3$ and $y = 2$

=> Fraction of rotten fruits = $\frac{2x+3y}{3x+4y}$

= $\frac{2(3)+3(2)}{3(3)+4(2)}$

= $\frac{12}{17}$

=> Ans - (A)

145. **C**

Principal amount for EMIs = Rs. (19,650 - 3,100) = Rs. 16,550

Rate of interest = 10% and number of EMIs = 3

Amount when compounded annually = $P(1 + \frac{r}{100})^T$

Let each installment = Rs. x

=> $\frac{x}{(1+\frac{10}{100})^1} + \frac{x}{(1+\frac{10}{100})^2} + \frac{x}{(1+\frac{10}{100})^3} = 16550$

=> $x[(\frac{10}{11}) + (\frac{100}{121}) + (\frac{1000}{1331})] = 16550$

=> $x[\frac{1210+1100+1000}{1331}] = 16550$

=> $x = 16550 \times \frac{1331}{3310}$

=> $x = Rs. 6,655$

=> Ans - (C)

146. **A**

The person bought the n article, each at Rs. d.

In that, he sold two articles at Rs. $\frac{d}{2}$, which brings in Rs. d for him.

Now, he sold the remaining n-2 articles at Rs. (d+8), which brings in Rs. (n-2)(d+8)

His total profit is Rs.72. This means, (n-2)(d+8)+d-nd = 72, or nd+8n-2d-16-nd = 72.

Hence 8n - 2d = 88, or 4n - d = 44

Now, d has to be greater than zero. Hence, the minimum value of n shall be 12 to make d as 4.

147. **C**

Dimensions of cube = 3 cm x 3 cm x 3 cm and dimensions of square hole = 1 cm x 1 cm

Outer surface area of 1 face = $(3 \times 3) - (1 \times 1) = 8\ cm^2$

=> Outer surface area of the cube (6 faces) = $6 \times 8 = 48\ cm^2$

Inner surface area of three tunnels = $3 \times 2[(2 \times 1) + (2 \times 1)] = 24\ cm^2$

$\therefore$ Total surface area of solid = $48 + 24 = 72\ cm^2$

=> Ans - (C)

148. **C**

The square will cover the same area of the triangle as much as covered by the triangle on square, and that at this point the amount of triangle covered is the same as the amount of square covered.

Let area of triangle be A and area of square = $6 \times 6 = 36\ cm^2$

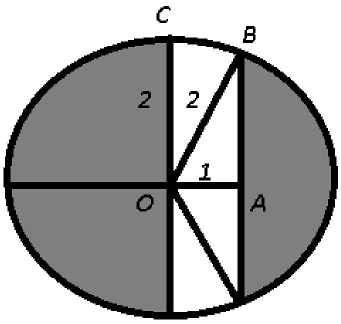
According to ques,

=> $\frac{60}{100} A = \frac{2}{3} \times 36$

=> $A = \frac{24}{0.6} = 40\ cm^2$

=> Ans - (C)

149. **D**



Radius of circle = OC = OB = 2 m and OA = 1 m

Area of path = $2 \times$[Area of $\triangle$ OAB+Area of sector BOC] ----------(i)

In right $\triangle$ AOB,

=> $(AB)^2 = (OB)^2 - (OA)^2$

=> $(AB)^2 = 4 - 1 = 3$

=> $(AB) = \sqrt{3}$ m ------------(ii)

Also, $tan(\angle\ AOB) = \frac{AB}{OA} = \sqrt{3}$

=> $\angle OAB = 60^\circ$

Thus, $\angle BOC = 90^\circ - 60^\circ = 30^\circ$ --------------(iii)

Now, in equation (i), area of path = $2 \times [(\frac{1}{2} \times OA \times AB) + (\frac{30^\circ}{360^\circ} \pi r^2)]$

= $2 \times [(\frac{1}{2} \times 1 \times \sqrt{3}) + (\frac{1}{12} \times \pi (2)^2)]$

= $2 \times [\frac{\sqrt{3}}{2} + \frac{\pi}{3}]$

= $\sqrt{3} + \frac{2\pi}{3}$ -----------(iv)

$\therefore$ Area of plot = Area of shaded region = Area of circle - Area of path

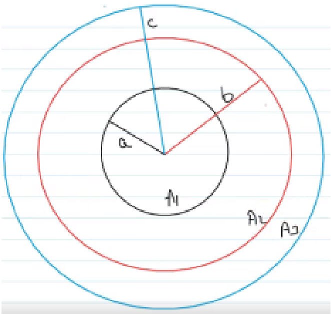
= $\pi (2)^2 - (\sqrt{3} + \frac{2\pi}{3})$

= $4\pi - \sqrt{3} - \frac{2\pi}{3}$

= $\frac{10\pi}{3} - \sqrt{3}\, m^2$

=> Ans - (D)

150. **A**



It is given that $a = 8$ cm and $b = 9$ cm. Also, $A_2 = \frac{A_1 + A_3}{2}$ ------------(i)

To find : $c = ?$

Solution : Area of a circle = πr^2

Using equation (i),

=> $\pi (b)^2 = \frac{\pi (a)^2 + \pi (c)^2}{2}$

=> $a^2 + c^2 = 2b^2$

=> $64 + c^2 = 162$

=> $c^2 = 162 - 64 = 98$

=> $c = \sqrt{98} = 7\sqrt{2}$ cm

=> Ans - (A)

151. **C**

Let us say that the biggest sphere (S1) has a diameter of $500\sqrt{3}$, which will be the diagonal of the cube (C1) fitted inside it. Hence, the side of the cube is of 500 cm.

Now, the side of C1 (500 cm) will be the diameter of another sphere (S2) fitted in it.

In S2, a cube (C2) is fitted which shall have a diagonal equal to the diameter of S2.

This means, the side of C2 is $\frac{500}{\sqrt{3}}$ cm. (as $a\sqrt{3}$ = diameter).

Now, the ratio of volume of C1 to volume of C2 will be $\frac{(500\sqrt{3})^3}{(\frac{500}{\sqrt{3}})^3}$, which is $3\sqrt{3} : 1$

152. **A**

Let the dimensions of rectangular box be $3, 4, 5$ units

=> Volume = $V = 3 \times 4 \times 5 = 60$ cu units

Case 1 : New length = 4 units and breadth and height remains same

=> $V' = 4 \times 4 \times 5 = 80$ cu units

Case 2 : New breadth = 5 units and length and height remains same

=> $V' = 3 \times 5 \times 5 = 75$ cu units

Case 3 : New height = 6 units length and breadth remains same

=> $V' = 3 \times 4 \times 6 = 72$ cu units

Clearly, increase in volume is greatest when x is increased.

=> Ans - (A)

Explanation [153 - 156]:

The number of students scoring more than 60% marks in school A = $\frac{55}{100} \times 800 = 440$

Thus, students scoring exactly 60% marks in school A = $800 - 440 - 320 = 40$

Similarly, for other schools :

School	< 60%	60%	>60%	Total
A	320	40	440	800
B	220	20	160	400
C	300	0	75	375
D	280	35	35	350
E	210	15	75	300

153. **A**

Clearly, each school but A, has more than half the students scoring less than 60% marks. Thus, **school A** has the required lowest percentage (400%)

=> Ans - (A)

154. **C**

Now, in schools A, B and E, there are **5%** of students scoring exactly 60% marks.

=> Ans - (C)

155. **D**

Total number of students scoring 60% or more marks is = (40+20+35+15) + (440+160+75+35+75)

= 110 + 785 = **895**

=> Ans - (D)

156. **B**

The number of students scoring more than 60% marks in school A = $\frac{55}{100} \times 800 = 440$

Thus, students scoring exactly 60% marks in school A = $800 - 440 - 320 = 40$

Similarly, for other schools :

School	< 60%	60%	>60%	Total
A	320	40	440	800
B	220	20	160	400
C	300	0	75	375
D	280	35	35	350
E	210	15	75	300

Total number of students scoring less than 60% marks = 320+220+300+280+210 = 1330

Total number of students scoring exactly 60% marks = 40+20+35+15 = 110

$\therefore$ Required ratio = $\frac{1330}{110} = 133 : 11$

=> Ans - (B)

157. **B**

Income in 2011 = 425 and expenditure of :

(A) : 2006 and 2007 = 275 + 225 = 500

(B) : 2007 and 2008 = 225 + 200 = **425**

(C) : 2006 and 2008 = 275 + 200 = 475

(D) : 2007 and 2010 = 225 + 225 = 450

=> Ans - (B)

158. **B**

Total income in the given years = 325+300+250+300+275+425 = 1875

=> Average income = $\frac{1875}{6} = 312.5$

$\therefore$ In two years, 2006 and 2011, income was more than average income.

=> Ans - (B)

159. **D**

Expenditure in 2010 = 225 and in 2011 = 325

=> Percentage increase = $\frac{325-225}{225} \times 100 = 44.4\%$

=> Ans - (D)

160. **C**

Ratio of expenditure to income :

(A) : 2009 = $\frac{300}{275} = 1.09$

(B) : 2008 = $\frac{250}{200} = 1.25$

(C) : 2007 = $\frac{300}{225} = 1.33$ **(max)**

(D) : 2006 = $\frac{325}{275} = 1.18$

=> Ans - (C)